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AUTOSAR HIGH-LEVEL DESIGN SPECIFICATION (HLD)

Document ID: HLD-BMS-2026-V1.0

System: Electric Vehicle Battery Management System (BMS)

Version: 1.0 (Baseline Architecture)

Classification: ASIL-C / ASIL-D

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1. System Architectural Overview

The Battery Management System (BMS) Software Architecture is designed in compliance with AUTOSAR 4.3 Classic Platform standards.

The BMS ECU monitors battery cells, executes state of charge (SOC) algorithms, manages high-voltage contactors, controls thermal management, and provides status information to the vehicle control system.

Primary Requirements:

- REQ_BMS_001: The system shall monitor all individual battery cell voltages with an update rate of 50ms.
- REQ_BMS_002: The system shall monitor cell temperatures and trigger cooling when temp exceeds 45°C.
- REQ_BMS_003: The system shall calculate Battery State of Charge (SOC) and State of Health (SOH).
- REQ_BMS_004: The contactor control manager shall open HV contactors upon critical over-current or isolation fault.
- REQ_BMS_005: Thermal management service shall regulate coolant pump flow based on temperature gradients.

2. Software Components (SWC) Definition

The application layer comprises the following core AUTOSAR Software Components:

- CellVoltageMonitor (SensorActuatorSWC):
Acquires cell voltages via ADC / SPI abstraction, performs low-pass filtering, and publishes cell voltage arrays.
- CellTemperatureMonitor (SensorActuatorSWC):
Reads NTC thermistor sensor arrays across battery modules, detects temperature hot spots.
- SOC_SOE_Estimator (ApplicationSWC):
Executes Extended Kalman Filter (EKF) algorithms to estimate State of Charge (SOC), State of Energy (SOE), and available power.
- ContactorControlManager (ApplicationSWC):
Implements the high-voltage precharge sequence, main contactor closing/opening logic, and emergency pyrofuse activation.
- ThermalManagementService (ServiceSWC):
Computes required cooling/heating thermal power and commands the chiller and coolant pump actuators.

3. Port Interfaces Specification

3.1 Sender-Receiver Interfaces

- If_CellVoltage:
Carries filtered cell voltage data elements [Array of uint16, mV].
Related Component: CellVoltageMonitor.

- If_CellTemperature:
Carries module temperature readings [Array of sint16, 0.1 deg C].
Related Component: CellTemperatureMonitor.

- If_BatteryStateOfCharge:
Carries calculated SOC percentage [uint16, 0.01% resolution] and SOE [kWh].
Related Component: SOC_SOE_Estimator.

- If_ThermalCommand:
Transmits cooling pump RPM and chiller valve position targets.
Related Component: ThermalManagementService.

3.2 Client-Server Interfaces

- If_ContactorCommand:
Operations: RequestPrecharge(), OpenMainRelay(), EmergencyShutdown().
Related Component: ContactorControlManager.

4. Component Port Allocations

- CellVoltageMonitor:
 - Port P_CellVoltage: Provided Port (PPort), Interface: If_CellVoltage

- CellTemperatureMonitor:
 - Port P_CellTemp: Provided Port (PPort), Interface: If_CellTemperature

- SOC_SOE_Estimator:
 - Port R_CellVoltage: Required Port (RPort), Interface: If_CellVoltage
 - Port R_CellTemp: Required Port (RPort), Interface: If_CellTemperature
 - Port P_BatteryStateOfCharge: Provided Port (PPort), Interface: If_BatteryStateOfCharge

- ContactorControlManager:
 - Port R_BatterySOC: Required Port (RPort), Interface: If_BatteryStateOfCharge
 - Port P_ContactorControl: Provided Port (PPort), Interface: If_ContactorCommand

- ThermalManagementService:
 - Port R_CellTemp: Required Port (RPort), Interface: If_CellTemperature
 - Port P_ThermalCmd: Provided Port (PPort), Interface: If_ThermalCommand